Oluwatosin Ayeni

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Cohere, London, UK

Dear Hiring Team,

I am applying for the Staff Software Engineer, Inference Infrastructure role at Cohere. I build and operate production LLM serving platforms at scale — Kubernetes, GPU fleets, multi-model routing, and full observability — and Cohere's focus on enterprise-grade open model deployment at the infrastructure layer is exactly the problem space I want to work on next.

Three specific achievements directly relevant to this role:

- **Production LLM serving platform on GKE:** I designed and operated *llm-inference-platform* — a full production stack on GKE (NVIDIA L4 24 GB, us-central1) running vLLM 0.32.0 and SGLang 0.5.20 serving Qwen3 8B AWQ and Llama 3.1 8B AWQ. LiteLLM Proxy 1.40.0 gateway with least-busy routing, A/B weighted round-robin, Redis response cache, and PostgreSQL spend tracking. KEDA autoscaling on GPU queue depth (not CPU — GPU pods at 5–15% CPU when saturated). Argo CD GitOps (3-min sync) + Argo Rollouts canary. Full Terraform IaC. Served 723 live requests, 1.25M+ tokens, $152.82 infra over 5.3 days.
- **GPU infrastructure at scale — H100 benchmarking:** I benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), serving Qwen3-8B BF16 at TP=2. Measured TTFT, TPOT, and throughput across concurrency 1–128. Peak: vLLM 5,965 tok/s. RadixAttention prefix caching delivered a 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context — 100% SLO compliance throughout. Live: inference-web-theta.vercel.app/benchmarks/llm.
- **Full observability and reliability stack:** 27-target Prometheus stack + Grafana 13 (3 custom dashboards: engine comparison, GPU utilisation, gateway) + AlertManager (6 rules) + Loki + FluentBit DaemonSet + NVIDIA DCGM Exporter via custom PodMonitor. NetworkPolicy default-deny + Kyverno pod security. Per-node hostPath weight cache (cold: ~5 min → cached: ~30 s).

Cohere occupies a rare position: building the enterprise LLM infrastructure layer that makes open models viable at production scale — serving commercial workloads with the reliability, latency, and cost guarantees that hyperscalers charge a premium for. The inference infrastructure team's challenge of running efficient, multi-tenant, low-latency serving across Cohere's model family is the class of problem I have been building towards independently. I am particularly drawn to working where inference efficiency is a core business priority, not an afterthought.

I am a focused, independent engineer who ships end to end: from Terraform and Kubernetes YAML to CUDA profiling to live dashboards. I look forward to discussing this role.

Yours sincerely,

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